W10L02

April 25, 2024

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TIME COMPLEXITY

Suppose $f(n) = 3n^2 - 2n + 5$ + trig

$f(n) = O(n^2)$

$$3n^2 - \underbrace{2n+5} \leq \overset{c}{c}\, n^2 \qquad \forall n \geq n_0$$

$f(n) = \sin n$ $\qquad$ n is radians

$f(n) = O(1)$

$$p_n(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

$$\boxed{p_n(x) = O(x^n)}$$

7, 2^n, n^4, $2n^2$, $\log n$, $\frac{1}{n}$, $5n$

A B C D E F G

$\underline{1/n}$, $\underline{7}$, $\underline{\log n}$, $\underline{5n}$, $\underline{2n^2}$, $\underline{n^4}$, $\underline{2^n}$

1 2 3 4 5 6 7

$\frac{1}{n} = O(7)$, $7 = O(\log n)$, $\log n = O(5n)$

$O(\text{constant} \ldots)$

$O(n) = O(5n) = O(n/2) \ldots$

Definition.

Let f, and g be two functions.

$$f, g : \mathbb{N} \to \mathbb{R}^+$$

We say " $f(n) = o(g(n))$ "

" f of n " is in " small oh of g of n "

iff

$$\lim_{n \to \infty} \frac{f(n)}{g(n)} = 0.$$

In other words, $f(n) = o(g(n))$

for any real number $c > 0$, number n_0 exists, s.t. $f(n) < c \cdot g(n) \quad \forall n \geq n_0$

$$\boxed{f(n) = o(g(n)) \iff \forall c > 0,\ c \in \mathbb{R},\ \exists n_0.\ f(n) < c \cdot g(n).\ \forall n \geq n_0}$$

$$f(n) = O(g(n)) \iff \exists c > 0,\ c \in \mathbb{R},\ \exists n_0,\ n_0 \in \mathbb{R}\quad f(n) \leq c \cdot g(n).\ \forall n \geq n_0$$

$f(n) = 3n^2 - 2n + 5$

$f(n) = O(n^2)$

$f(n) \neq o(n^2)$, $f(n) = o(n^3)$

$A = \{0^k 1^k \mid k \geq 0\}$ 0010 X

M_1 = "On input string w:"

$O(n)$ 1. Scan accross the tape and reject if there is a 0 on the right of 1.

$n/2$ step $= O(n)$ 2. Repeat if both 0's & 1's remain on the tape:

3. Scan accross the tape, crossing off a single 0 and a single 1.

4. If 0's sill remain after all 1's have crossed off, or if 1's remain after all 0's crossed off, reject

5. If none left accept."

$\cancel{0}\cancel{0}\cancel{0}\cancel{1}\cancel{1}\cancel{1}\sqcup$

$\cancel{0}\cancel{0}\cancel{0}\cancel{1}\cancel{1}\cancel{1}1\sqcup$ ✗

$\underbrace{\cancel{0}0\cdots0\cancel{1}\cdots1}_{n}\sqcup \in \underbrace{A}$ ✓

✓ $\underbrace{O(n)}$ check the format

$n+n$ $\underline{2n}$

n

$n-2$ $\frac{n}{\cancel{2}}\cdot\cancel{2}n = n^2$

$n-4$

$\underbrace{O(n)}+\underbrace{O(n^2)} = \underline{\underline{O(n^2)}}$

$O(n^2)$ algorithm to decide A.

Definition:

Let $t: \mathbb{N} \to \mathbb{R}^+$ be a fuction.

We define the "time complexity class"

$\mathrm{TIME}(t(n))$, to be the collection

of all languages that are

decidable by an $O(t(n))$ time

Turing-machine.

Example:

$$\mathrm{TIME}(n)$$

$$\mathrm{TIME}(n^2)$$

Class P

$$P = \bigcup_{\substack{n \geq 0 \\ n \in \mathbb{N}}} \mathrm{TIME}(n^k)$$

Class NP

$$NP = \bigcup_{\substack{n \geq 0 \\ n \in \mathbb{N}}} \mathrm{NTIME}(n^k)$$

is $P = NP$?

$\cancel{0}\ 0\ \cancel{0}\ 0\ \cancel{0}\ \cancel{1}\ 1\ \cancel{1}\ 1\ \cancel{1}$

$\cancel{0}\ 0\ \cancel{1}\ 1$

$\cancel{0}\ \cancel{1}$

$$\underbrace{0 \cdots 0\, 1 \cdots 1}_{n}$$

$\lfloor n/2 \rfloor$

$\lfloor n/2^2 \rfloor$

$\vdots$

$n = 2^k \implies k = \log_2 n$ $\quad \lfloor n/2^k \rfloor$

$k+1 = O(\log n)$ $\quad \lfloor n/2^{k+1} \rfloor = 0$